



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 12

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COMMON QUARTERLY EXAMINATION - 2024-25 BUSINESS MATHEMATICS AND STATISTICS

Time Allowed : 3.00 Hours]

PART - A

[Max. Marks : 90

I. Answer All the questions.

20x1=20

Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer.

1. The rank of the unit matrix of order n is

- (a) $n - 1$ (b) n (c) $n + 1$ (d) n^2

2. The rank of $m \times n$ matrix whose elements are unity is

- (a) 0 (b) 1 (c) m (d) n

3. The system of equations $4x + 6y = 5$, $6x + 9y = 7$ has

- (a) a unique solution (b) no solution (c) infinitely many solutions (d) none of these

4. If $\frac{a_1}{x} + \frac{b_1}{y} = c_1$, $\frac{a_2}{x} + \frac{b_2}{y} = c_2$, $\Delta_1 = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$; $\Delta_2 = \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}$; $\Delta_3 = \begin{vmatrix} c_1 & a_1 \\ c_2 & a_2 \end{vmatrix}$ then (x, y)

- (a) $\left(\frac{\Delta_2}{\Delta_1}, \frac{\Delta_3}{\Delta_1}\right)$ (b) $\left(\frac{\Delta_3}{\Delta_1}, \frac{\Delta_2}{\Delta_1}\right)$ (c) $\left(\frac{\Delta_1}{\Delta_2}, \frac{\Delta_1}{\Delta_3}\right)$ (d) $\left(\frac{-\Delta_1}{\Delta_2}, \frac{-\Delta_1}{\Delta_3}\right)$

5. $\int e^{2x} [2x^2 + 2x] dx$

- (a) $e^{2x} x^2 + c$ (b) $x e^{2x} + c$ (c) $2x^2 e^2 + c$ (d) $\frac{x^2 e^x}{2} + c$

6. $\int \frac{2x+3}{\sqrt{x^2+3x+2}} dx$ is

- (a) $\sqrt{x^2+3x+2} + c$ (b) $2\sqrt{x^2+3x+2} + c$ (c) $\log(x^2+3x+2) + c$ (d) $\frac{2}{3}(x^2+3x+2)^{\frac{3}{2}} + c$

7. If $n > 0$, then $\Gamma(n)$ is

- (a) $\int_0^1 e^{-x} x^{n-1} dx$ (b) $\int_0^1 e^{-x} x^n dx$ (c) $\int_0^\infty e^x x^{-n} dx$ (d) $\int_0^\infty e^{-x} x^{n-1} dx$

8. $\int \frac{\sin 2x}{2 \cos x} dx$ is

- (a) $\sin x + c$ (b) $\frac{1}{2} \cos x + c$ (c) $-\cos x + c$ (d) $\frac{1}{2} \cos x + c$

9. If MR and MC denotes the marginal revenue and marginal cost functions, then the profit functions is

- (a) $P = \int (MR - MC) dx + k$ (b) $P = \int (MR + MC) dx + k$
(c) $P = \int (MR)(MC) dx + k$ (d) $P = \int (R - C) dx + k$

10. If the marginal revenue $MR = 35 + 7x - 3x^2$, then the average revenue AR is

- (a) $35x + \frac{7x^2}{2} - x^3$ (b) $35 + \frac{7x}{2} - x^2$ (c) $35 + \frac{7x}{2} + x^2$ (d) $35 + 7x + x^2$

11. Area bounded by $y = e^x$ between the limits 0 to 1 is

- (a) $(e - 1)$ sq.units (b) $(e + 1)$ sq.units (c) $(1 - \frac{1}{e})$ sq.units (d) $(1 + \frac{1}{e})$ sq.units



12. The demand and supply function of a commodity are $P(x) = (x - 5)^2$ and $S(x) = x^2 + x + 3$ then the quantity x_0 is

- (a) 5 (b) 2 (c) 3 (d) 19

13. The order and degree of the differential equation $\sqrt{\frac{d^2y}{dx^2}} = \sqrt{\frac{dy}{dx} + 5}$ are respectively

- (a) 2 and 3 (b) 3 and 2 (c) 2 and 1 (d) 2 and 2

14. Solution of $\frac{dx}{dy} + Px = 0$

- (a) $x = ce^{Py}$ (b) $x = ce^{-Py}$ (c) $x = py + c$ (d) $x = cy$

15. The differential equation formed by eliminating A and B from $y = e^{-2x}(A \cos x + B \sin x)$ is

- (a) $y_2 - 4y_1 + 5 = 0$ (b) $y_2 + 4y - 5 = 0$ (c) $y_2 - 4y_1 - 5 = 0$ (d) $y_2 + 4y_1 + 5 = 0$

16. The integrating factor of $\cos^2 x \frac{dy}{dx} + y = \tan x$ is

- (a) $e^{\cos x}$ (b) $e^{\tan x}$ (c) $e^{\sin x}$ (d) e^x

17. If $h=1$, then $\Delta(x^2) =$

- (a) $2x$ (b) $2x - 1$ (c) $2x + 1$ (d) 1

18. If m and n are positive integers then $\Delta^m \Delta^n f(x) =$

- (a) $\Delta^{m+n} f(x)$ (b) $\Delta^m f(x)$ (c) $\Delta^n f(x)$ (d) $\Delta^{m-n} f(x)$

19. For the given data find the value of $\Delta^3 y_0$ is

x	5	6	9	11
y	12	13	15	18

- (a) 1 (b) 0 (c) 2 (d) -1

20. $\Delta^2 \Delta^3 f(x) =$

- (a) $\Delta^2 f(x)$ (b) $\Delta^3 f(x)$ (c) $\Delta^5 f(x)$ (d) $\Delta f(x)$

PART - II

1. Answer any 7 questions

2. Each question carries 2 marks

3. Question number 30 is compulsory

7x2=14

21. Find the rank of the matrix $\begin{pmatrix} -5 & -7 \\ 5 & 7 \end{pmatrix}$

22. Solve the following equations by using Cramer's rule : $2x + 3y = 7$; $3x + 5y = 9$

23. Evaluate the following: $\int_0^\infty e^{-mx} x^6 dx$

24. Integrate the following with respect to x: $\frac{1}{2x^2 - 9}$

25. Find the area bounded by the line $y = x$, the x-axis and the ordinates $x = 1$, $x = 2$.

26. The demand function for a commodity is $p = e^{-x}$. Find the consumer's surplus when $p = 0.5$.



27. Solve: $\frac{dy}{dx} - \frac{y}{x} = x$

28. Solve the differential equation: $\frac{d^2y}{dx^2} + 16y = 0$

29. If $f(x) = x^2 + 3x$ then show that $\Delta f(x) = 2x + 4$

30. Prove that $(1 + \Delta)(1 - \nabla) = 1$.

PART - III

7 x 3 = 21

1. Answer any 7 questions
2. Each question carries 3 marks
3. Question number 40 is compulsory

31. Find the rank of $\begin{pmatrix} 3 & 1 & -5 & -1 \\ 1 & -2 & 1 & -5 \\ 1 & 5 & -7 & 2 \end{pmatrix}$

32. The total cost of 11 pencils and 3 erasers is ₹ 64 and the total cost of 8 pencils and 3 erasers is ₹ 49. Find the cost of each pencil and each eraser by Cramer's rule.

33. Integrate the following with respect to x : $\frac{1}{\sqrt{x+1} + \sqrt{x-1}}$

34. Using second fundamental theorem, evaluate the following: $\int_1^e \frac{dx}{x(1+\log x)^3}$

35. Find the area of the parabola $y^2 = 8x$ bounded by its latus rectum.

36. An account fetches interest at the rate of 5% per annum compounded continuously. An individual deposits ₹1,000 each year in his account. How much will be in the account after 5 years. ($e^{0.25} = 1.284$).

37. Find the differential equation corresponding to $y = ae^{4x} + be^{-x}$ where a, b are arbitrary constants,

38. Solve: $(1-x)dy - (1+y)dx = 0$

39. Evaluate $\Delta \left[\frac{1}{(x+1)(x+2)} \right]$ by taking '1' as the interval of differencing.

40. A second degree polynomial passes through the point (1,-1) (2,-1) (3,1) (4,5). Find the polynomial.

PART - IV

1. Answer all the questions
2. Each question carries 5 marks

7 x 5 = 35

41. a) For what values of the parameter λ , will the following equations fail to have unique solution:

$3x - y + \lambda z = 1, 2x + y + z = 2, x + 2y - \lambda z = -1$ by rank method.

(OR)

b) The price of 3 Business Mathematics books, 2 Accountancy books and one Commerce book is ₹840. The price of 2 Business Mathematics books, one Accountancy book and one Commerce book is ₹570. The price of one Business Mathematics book, one Accountancy book and 2 Commerce books is ₹630. Find the cost of each book by using Cramer's rule.



42. a) An amount of ₹5,000/- is to be deposited in three different bonds bearing 6%, 7% and 8% per year respectively. Total annual income is ₹358/-. If the income from first two investments is ₹70/- more than the income from the third, then find the amount of investment in each bond by rank method.

(OR)

b) Evaluate $\int_2^5 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{7-x}} dx$

43. a) Evaluate the following integrals as the limit of the sum: $\int_1^3 (2x + 3) dx$

(OR)

b) Evaluate $\int \frac{3x+2}{(x-2)^2(x-3)} dx$

44. a) The marginal cost of production of a firm is given by $C'(x) = 5 + 0.13x$, the marginal revenue is given by $R'(x) = 18$ and the fixed cost is ₹ 120. Find the profit function.

(OR)

- b) A firm has the marginal revenue function given by $MR = \frac{a}{(x+b)^2} - c$ where x is the output and a, b, c are constants.

Show that the demand function is given by $x = \frac{a}{b(p+c)} - b$.

45. a) The demand equation for a products is $x = \sqrt{100 - p}$ and the supply equation is $x = \frac{p}{2} - 10$. Determine the consumer's surplus and producer's surplus, under market equilibrium.

(OR)

- b) Solve the differential equation $\frac{dy}{dx} = \frac{x-y}{x+y}$.

46. a) If $\frac{dy}{dx} + 2y \tan x = \sin x$ and if $y = 0$ when $x = \frac{\pi}{3}$ express y in terms of x .

(OR)

- b) Solve the differential equation $(D^2 + D - 6)y = e^{3x} + e^{-3x}$

47. a) Find the missing entries from the following.

X	0	1	2	3	4	5
y=f(x)	0	-	8	15	-	35

(OR)

- b) Find $f(2.8)$ from the following table.

X	0	1	2	3
f(x)	1	2	11	34